

Individual assignment 5 Test

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)

# colors
library(NatParksPalettes)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data",
  "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

Code from Class

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
```

```

# group by date_site column
group_by(date_site) |>
# calculate median pH, salinity, DO
summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter
# (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame

```

```

ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal,
                              .fun = "median", .na_rm = TRUE),
       site_full = fct_rev(site_full))

```

1. Visualizing differences across sites in major taxon abundance

a) Plan your figure

- x-axis: site_full
- y-axis: log_ave_lit
- geom for avg abundance: jitter
- geom for medians: point

b) Wrangle your data

```

# creating new object from clean aquatic inverts
ostracods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "ostracod") |>
# log + 1 transform mean abundance per liter

```

```
mutate(log_ave_lit = log(ave_lit + 1))

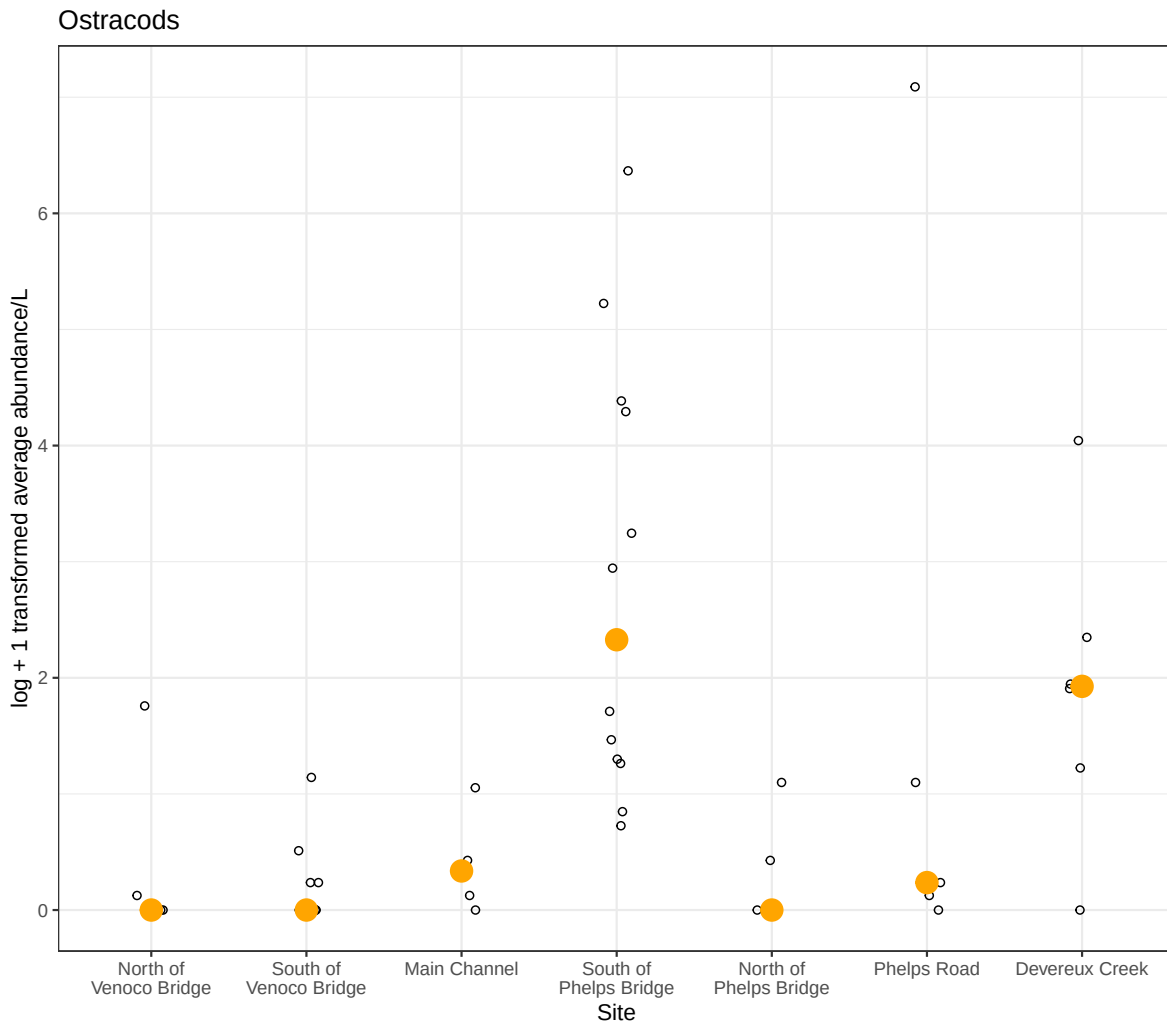
slice_sample(ostracods)

# A tibble: 1 x 24
  date_site      date                site  taxon  ave_lit med_ph med_sal med_do
  <chr>          <dtm>                <chr> <chr>    <dbl>  <dbl>  <dbl>  <dbl>
1 2023-05-16_NVBR 2023-05-16 00:00:00 NVBR  ostra~  0.133  9.29    NA    9.83
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c) Code your figure

```
# base layer
ostracodgraph <- ggplot(data = ostracods,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
  color = "orange",
  size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
  y = "log + 1 transformed average abundance/L",
  title = "Ostracods") +
# different theme
theme_bw()

# displaying object
ostracodgraph
```



d) Create a multipanel figure

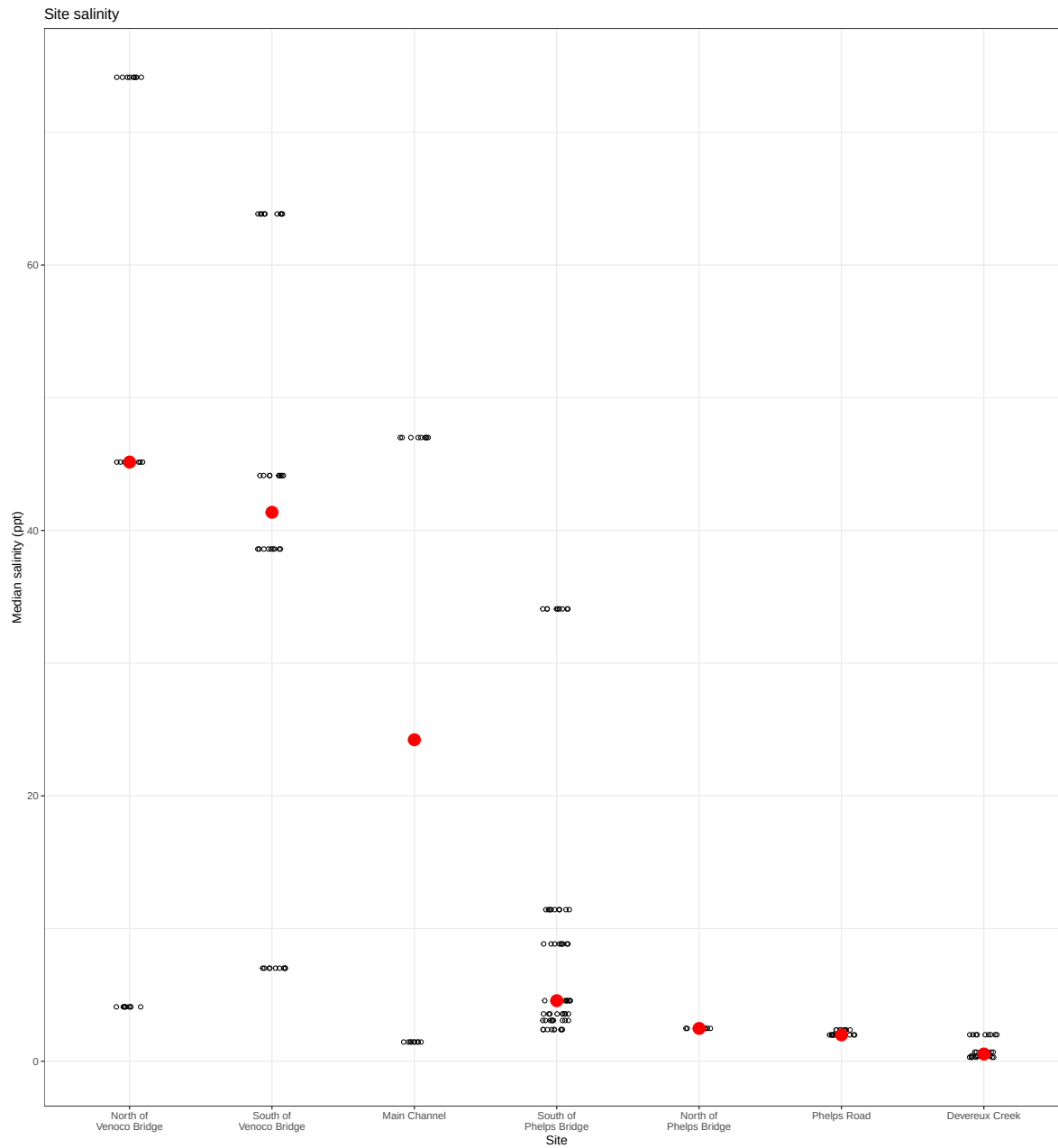
```
# salinity
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
  # first layer: salinity measurements at each survey
  geom_jitter(height = 0,
    width = 0.1,
```

```

        shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
              color = "red",
              size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site salinity") +
# different theme
theme_bw()

# displaying object
salinity

```



```
# cleaning copepods before graphing
copepods <- aq_ins_clean |>
  # filter to only include copepods
  filter(taxon == "copepod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
```

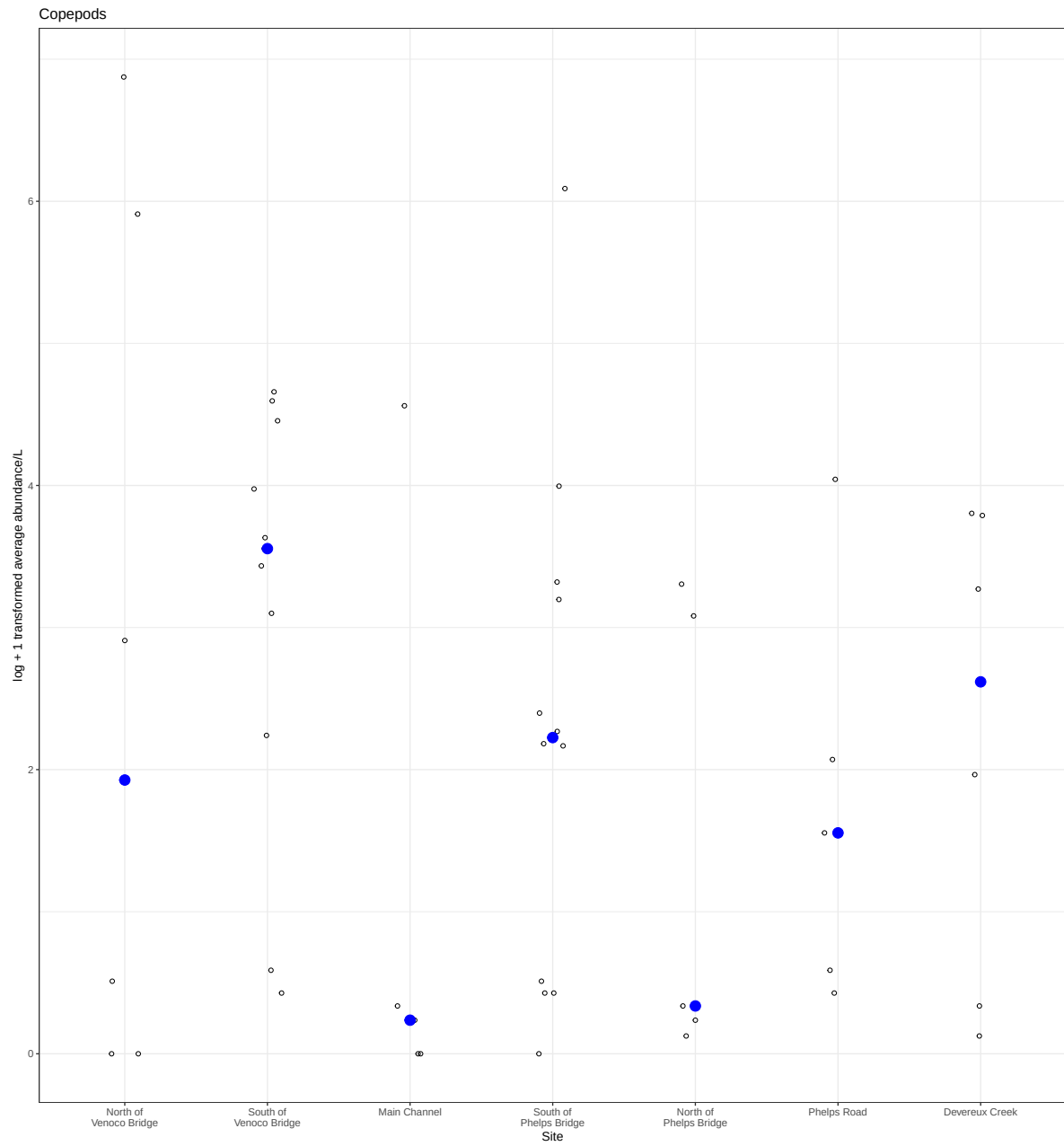


```

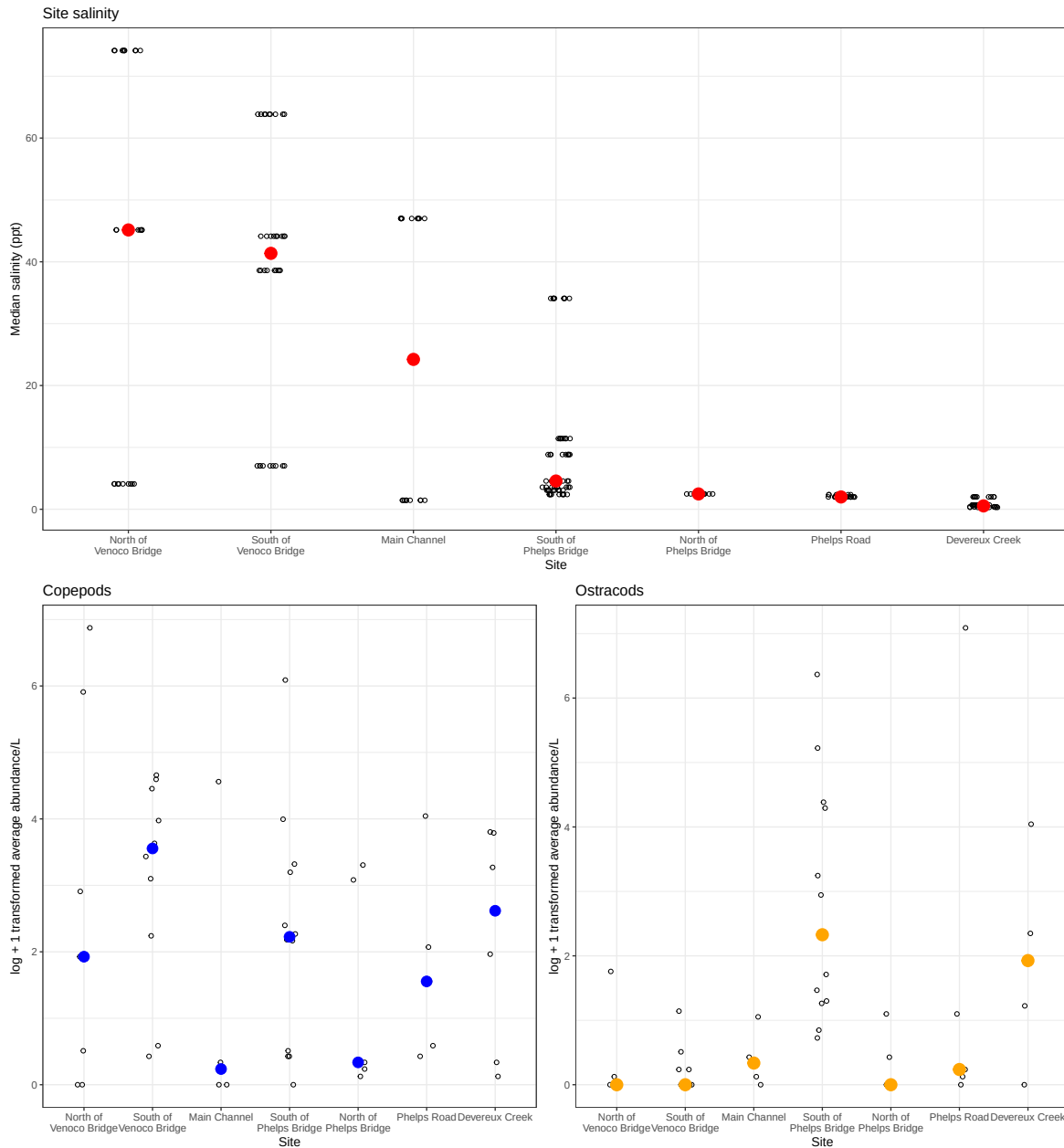
# base layer
copepods <- ggplot(data = copepods,
                   # x-axis: site
                   # y-axis: mean abundance/L
                   mapping = aes(x = site_full,
                                y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
            shape = 21,
            width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
            fun = "median",
            size = 4,
            color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepods") +
# different theme
theme_bw()

# displaying object
copepods

```



salinity / (copepods | ostracodgraph)



e) Interpret figure

- Copepods and ostracods had similar levels of abundance at the sites 'South of Phelps Bridge' and 'Devereaux Creek' with values being relatively high, and also had similar low values at 'Main Channel' and 'North of Phelps Bridge'. At the other sites, the values

differed, with Copepods having more abundance in all of the other sites than Ostracods.

- I do not think that ostracod abundance relates to site level differences in salinity, because sites with lower salinity produced high and low levels of ostracods (Devereux Creek, Phelps Road, North of Phelps Bridge, South of Phelps Bridge).

#2. Visualizing total abundance as a function of salinity

a) Plan your figure

- x-axis: med_sal
- y-axis: log_ave_lit
- color: scientificName
- geom: jitter
- facets: facet_wrap(~variable, scales = "free")

b) Wrangle your data

```
# creating new object from clean aquatic inverts
logsalinity2 <- aq_ins_clean |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

slice_sample(logsalinity2)
```

```
# A tibble: 1 x 24
  date_site      date                site  taxon    ave_lit med_ph med_sal med_do
  <chr>          <dtm>                <chr> <chr>    <dbl>  <dbl>  <dbl>  <dbl>
1 2022-01-21_NEC 2022-01-21 00:00:00 NEC    cladoc~      0    NA    NA    NA
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

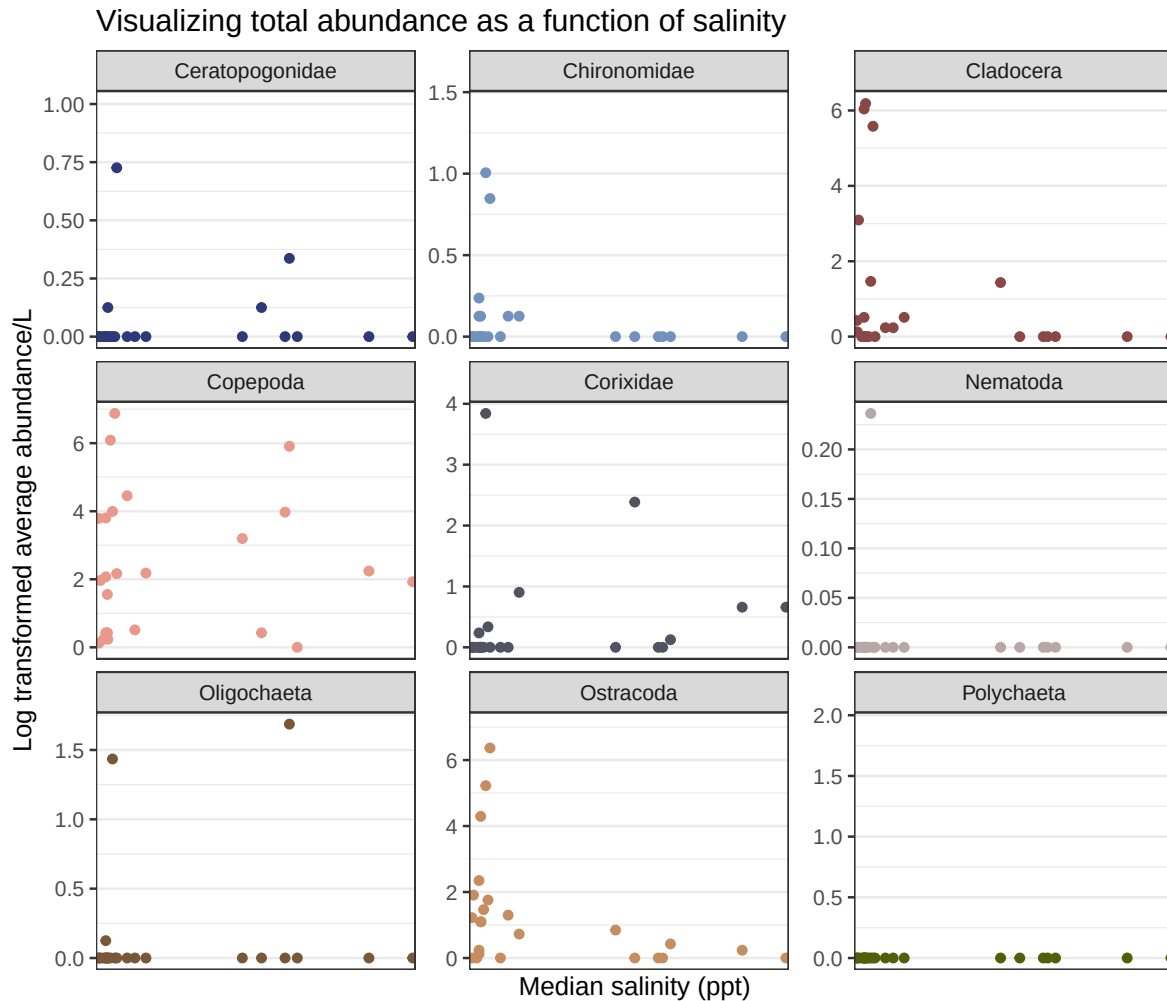
c) Code your figure

```

# base layer
logsal <- ggplot(data = logsalinity2,
  # x-axis: salinity
  # y-axis: log_ave_lit
  mapping = aes(x = med_sal,
    y = log_ave_lit,
    color = scientificName)) +
  geom_point() +
  # wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 15)) +
  # relabelling x- and y-axes
  labs(x = "Median salinity (ppt)",
    y = "Log transformed average abundance/L",
    title = "Visualizing total abundance as a function of salinity") +
  # different theme
  theme_bw() +
  facet_wrap(~scientificName, scales = "free") +
  scale_color_natparks_d("Torres") +
  theme(legend.position = "none")

# displaying object
logsal

```



d) Interpret your figure

The Ceratopogonidae, the Chironomidae, the Cladocera, the Corixidae, and the Ostracoda had similar relationships between salinity and abundance, with higher levels at low salinities, and lower levels at higher salinities. The Nematoda, Polychaeta, Copepoda, and Oligochaeta did not show a clear relationship between salinity and abundance. Most plots started with high levels of abundance with low salinities, including the Ceratopogonidae, Chironomidae, Cladocera, Copepoda, Corixidae, Nematoda, Oligochaeta, and Ostracoda.